Metrics for TWINX

A. Distance and path efficiency

- Actual distance travelled: sum of consecutive measured position changes.
- Start-to-goal distance: straight-line distance between the initial and final positions.
- Planned path length: length of the route generated by the planner.
- Path efficiency: straight-line distance divided by actual distance travelled.
- Path similarity: compare predicted and measured trajectories using a measure such as the Fréchet distance or average cross-track error.

B. Trajectory tracking and prediction error

- Mean position error: average distance between predicted and measured positions at corresponding times.
- RMSE: root-mean-square position error, which gives more weight to larger deviations.
- Final position error: distance between the final position and the goal.
- Cross-track error: distance from the robot's position to the planned path.
- Heading error: difference between predicted and measured robot orientation.

C. Travel time, velocity and motion behaviour

- Travel time: elapsed time from movement start to goal arrival.
- Average speed: distance travelled divided by elapsed time
- Peak speed: highest measured speed.
- Velocity tracking error: difference between predicted and measured velocity.
- Motion smoothness: acceleration variation or jerk, where appropriate.
- Stability: body roll, pitch and vertical movement during locomotion.

D. Localization accuracy

- Position localization error: estimated position compared with a reference or ground-truth position.
- Orientation error: estimated heading compared with ground truth.
- Localization drift: how position error changes over the route.

This requires a localization estimate and an independent reference position. In the current visualization, the script sets the robot's pose directly, so its displayed pose is not an independent localization measurement.

E. Obstacle avoidance and navigation success

- Minimum obstacle clearance: smallest measured separation between the robot and obstacle.
- Collision count: number of detected contacts or collisions with the obstacle.
- Obstacle clearance margin: clearance relative to the required safe distance.
- Avoidance deviation: how far the robot detours from the direct route.
- Navigation success rate: successful runs divided by total runs.
- Completion time and timeout rate: whether the goal is reached within the allowed time.

Collision and clearance metrics should use the robot's actual geometry, not just the distance between its base and the obstacle center.

F. Digital-twin prediction and calibration

- Prediction error before calibration: mismatch between the baseline prediction and observed motion.
- Prediction error after calibration: mismatch after updating selected model parameters.
- Improvement percentage: reduction in error after calibration.
- Repeatability: variation in results across repeated runs under the same conditions.

- Parameter sensitivity: how strongly selected parameters affect prediction accuracy.